



In Class II, we have learnt how to add two or more 2-digit and 3-digit numbers. We shall now extend the same idea add to 4-digit numbers.

Addends and the sum: When two or more numbers are added, each of the numbers to be added is known as an **addend** and the result obtained after addition is called the **sum**.

Addition without Carrying



How to Add?

Step 1: Arrange the digits of the given numbers in columns of thousands, hundreds, tens and ones.

Step 2: Add columnwise.

First add the ones, then add the tens, followed by the hundreds and finally add the thousands.



Solved Examples

Example 1: Add 5461 and 2137.

Solution: Arranging the given numbers in columns and adding columnwise, we get:

	Th	H	T	O
	5	4	6	1
+	2	1	3	7
	7	5	9	8

Explanation:

- Adding ones** : 1 one + 7 ones = 8 ones
Write 8 under ones column.
- Adding tens** : 6 tens + 3 tens = 9 tens
Write 9 under tens column.
- Adding hundreds** : 4 hundreds + 1 hundred = 5 hundreds
Write 5 under hundreds column.
- Adding thousands** : 5 thousands + 2 thousands = 7 thousands
Write 7 under thousands column.

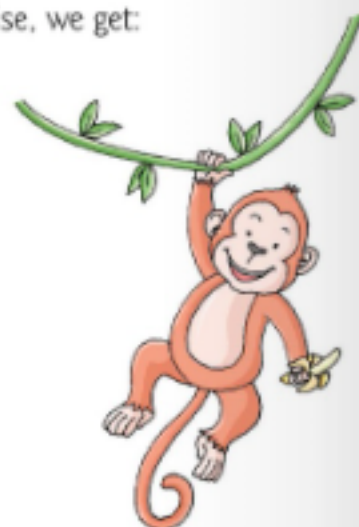
Hence, the sum of the given numbers is 7598.



Example 2: Find the sum: $5142 + 1203 + 2631$.

Solution: Arranging the given numbers in columns and adding columnwise, we get:

	Th	H	T	O
	5	1	4	2
	1	2	0	3
+	2	6	3	1
	8	9	7	6



Explanation:

Adding ones : $2 + 3 + 1 = 6$. Write 6 under ones column.

Adding tens : $4 + 0 + 3 = 7$. Write 7 under tens column.

Adding hundreds : $1 + 2 + 6 = 9$. Write 9 under hundreds column.

Adding thousands : $5 + 1 + 2 = 8$. Write 8 under thousands column.

Hence, the sum of the given numbers is 8976.



Exercise 6

Find the sum.

1.

Th	H	T	O	
8	5	3	2	
+	1	3	4	6

2.

Th	H	T	O	
4	9	2	6	
+	5	0	6	3

3.

Th	H	T	O	
6	4	7	5	
+	3	5	1	3

4.

Th	H	T	O	
7	2	5	4	
+	2	6	3	5

5.

Th	H	T	O	
4	5	3	6	
+	1	3	4	2

6.

Th	H	T	O	
3	5	4	7	
+	5	3	5	2

7.

Th	H	T	O
5	4	2	1
+	4	3	6

8.

Th	H	T	O
6	3	0	2
+	2	4	7



Add the following.

9.

Th	H	T	O
1	2	4	3
4	6	3	5
+	1	0	1

10.

Th	H	T	O
2	3	5	4
1	0	2	3
+	5	1	2

11.

Th	H	T	O
3	4	2	1
4	2	5	3
+	3	1	4

12.

Th	H	T	O
6	5	2	3
1	2	5	4
+	2	1	2

13.

Th	H	T	O
3	2	1	4
2	3	2	1
+	2	6	3

14.

Th	H	T	O
5	3	2	4
	6	3	0
+		4	5



15.

Th	H	T	O
5	6	3	2
		5	4
+	3	0	2

16.

Th	H	T	O
7	8	0	6
	1	6	1
+		3	2

Arrange in columns and find the sum.

17. $2435 + 132$

19. $6241 + 1326 + 432$

Arrange in columns and add.

21. 3562 and 4326

23. 6381 , 2102 and 1000

18. $6230 + 356 + 13$

20. $2401 + 1036 + 101 + 60$

22. 1405 , 231 , 42 and 11

24. 5360 , 407 , 111 and 21

Addition with Carrying

The concept will be clear from the following examples.



Solved Examples

Example 1: Add 4378 and 3764.

Solution: Arranging the given numbers in columns and adding columnwise, we get:

	Th	H	T	O
	(1)	(1)	(1)	← Carry
	4	3	7	8
+	3	7	6	4
	8	1	4	2



Explanation:

Adding ones

$$\begin{aligned} &: 8 \text{ ones} + 4 \text{ ones} = 12 \text{ ones} \\ &= 10 \text{ ones} + 2 \text{ ones} \\ &= 1 \text{ ten} + 2 \text{ ones.} \end{aligned}$$

Write 2 under ones column.

Carry over 1 ten to tens column.

Adding tens

$$\begin{aligned} &: 1 \text{ ten (carried over)} + 7 \text{ tens} + 6 \text{ tens} = 14 \text{ tens} \\ &= 10 \text{ tens} + 4 \text{ tens} \\ &= 1 \text{ hundred} + 4 \text{ tens.} \end{aligned}$$

Write 4 under tens column.

Carry over 1 hundred to hundreds column.

Adding hundreds

$$\begin{aligned} &: 1 \text{ hundred (carried over)} + 3 \text{ hundreds} + 7 \text{ hundreds} \\ &= 11 \text{ hundreds} \\ &= 10 \text{ hundreds} + 1 \text{ hundred} \\ &= 1 \text{ thousand} + 1 \text{ hundred} \end{aligned}$$

Write 1 under hundreds column.

Carry over 1 thousand to thousands column.

Adding thousands : 1 thousand (carried over) + 4 thousands + 3 thousands = 8 thousands.

Write 8 under thousands column.

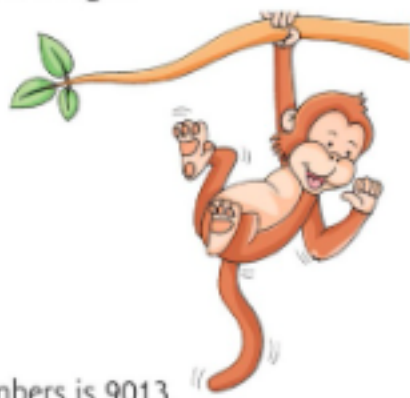
Hence, the sum of the given numbers is 8142.

Example 2: Find the sum: 4576 + 2784 + 1653.

Solution: Arranging the given numbers in columns and adding columnwise, we get:

	Th	H	T	O
	2	2	1	← Carry
	4	5	7	6
	2	7	8	4
+	1	6	5	3
	9	0	1	3

Hence, the sum of the given numbers is 9013.



Example 3: Find the sum: 5378 + 689 + 204 + 67.

Solution: Arranging the given numbers in columns and adding columnwise, we get:

	Th	H	T	O
	1	2	2	← Carry
	5	3	7	8
		6	8	9
		2	0	4
+			6	7
	6	3	3	8

Hence, the sum of the given numbers is 6338.



Exercise 7

Find the sum.

1.

	Th	H	T	O
	7	6	3	5
+	1	3	6	5

2.

	Th	H	T	O
	7	9	8	9
+	1	1	1	1

3.

	Th	H	T	O
	3	7	0	8
+	4	6	0	9

4.

Th	H	T	O
3	5	6	8
5	8	9	5

5.

Th	H	T	O
4	7	6	9
2	5	8	7

6.

Th	H	T	O
5	9	8	9
2	3	6	4

7.

Th	H	T	O
6	2	7	9
2	9	4	6

8.

Th	H	T	O
5	8	6	7
3	6	5	4



Add the following.

9.

2	7	5	8
3	4	6	2
1	3	4	5

10.

3	5	6	7
1	9	9	4
4	2	3	5

11.

4	6	7	8
2	9	6	5
1	3	5	7

12.

6	3	5	7
2	5	4	9
	6	7	8

13.

5	8	6	9
	3	7	4
3	5	8	6

14.

6	8	7	9
	4	5	6
	7	3	4



15.

6	9	8	4
		2	9
	5	3	6
	2	0	8

16.

7	1	6	5
	6	7	8
		4	6
1	0	5	3

Find the sum.

17. $1378 + 2659 + 1235$

19. $1998 + 2146 + 99 + 9$

Arrange in columns and add.

21. 3047, 2964 and 3782

23. 3874, 487, 96 and 1065

Fill in the missing numerals.

25.

	4		3
6		6	
9	0	2	1

18. $4059 + 816 + 1977$

20. $3276 + 689 + 78 + 127$

22. 2165, 3978, 459 and 78

24. 8888, 777, 66, 155 and 8

26.

5		8	
2	3		9
	1	6	6

27.

1	3		
		6	8
9	3	1	7

28.

6			4
1	7	9	
	0	8	2

29.

	2		7
1	8	9	
5		8	3



Properties of Addition

I. Order Property or Commutative Property for Addition

Find the sum: $3265 + 2678$.

Also, find the sum: $2678 + 3265$.

What do you conclude?

We have:

	1	1	← Carry
3	2	6	5
+	2	6	7 8
5	9	4	3

and

	1	1	← Carry
2	6	7	8
+	3	2	6 5
5	9	4	3

$$\therefore 3265 + 2678 = 5943.$$

$$\text{And, } 2678 + 3265 = 5943.$$

Conclusion: $3265 + 2678 = 2678 + 3265$.

The sum of two numbers does not change when the order of the addends is changed.

This is called the **order property** or **commutative property** for addition of numbers.

II. Grouping Property or Associative Property for Addition

Find the sum: $(1368 + 2452) + 1073$.

Also, find the sum: $1368 + (2452 + 1073)$.

What do you conclude?

We have:

$$1368 + 2452 = 3820$$

$$\therefore (1368 + 2452) + 1073$$

$$= 3820 + 1073$$

$$= 4893.$$

$$\text{And, } 2452 + 1073 = 3525$$

$$\therefore 1368 + (2452 + 1073)$$

$$= 1368 + 3525$$

$$= 4893.$$

	1	1	← Carry
1	3	6	8
+	2	4	5 2
3	8	2	0

3	8	2	0
+	1	0	7 3
4	8	9	3

	1	← Carry
2	4	5 2
+	1	0 7 3
3	5	2 5

Carry →	1		
1	3	6	8
+	3	5	2 5
4	8	9	3

Conclusion: $(1368 + 2452) + 1073 = 1368 + (2452 + 1073)$.

The sum of three or more numbers does not change even when their grouping is changed.

This is known as the **grouping property** or **associative property** for addition of numbers.



III. Additive Property for Zero

Find the sums: $1674 + 0$ and $0 + 1674$.

Th	H	T	O
1	6	7	4
			0
1	6	7	4

and

Th	H	T	O	
			0	
+	1	6	7	4
	1	6	7	4

We have:

$$\therefore 1674 + 0 = 1674 \text{ and } 0 + 1674 = 1674.$$

The sum of a number and 0 is the number itself.

This is called the **additive property** of 0.



Exercise 8

Fill in the blanks.

- $2357 + 1876 = 1876 + \dots$
- $4689 + 1831 = \dots + 4689$
- $\dots + 2391 = 2391 + 5137$
- $1064 + \dots = 6397 + 1064$
- $\dots + 3716 = 3716$
- $5965 + \dots = 5965$
- $(1008 + 999) + 1066 = \dots + (999 + 1066)$
- $1938 + (2346 + 1650) = (1938 + 2346) + \dots$
- $(2346 + 1530) + 1734 = \dots + (1530 + 1734)$
- $(4375 + 1625) + 1595 = (1625 + \dots) + 1595$
- Find the sum by suitable grouping.**
 - $53 + 42 + 47 + 58$
 - $44 + 34 + 56 + 66$
 - $291 + 378 + 109 + 122$



Word Problems

Example 1: Mr Khanna purchased a washing machine for ₹ 5468 and a geyser for ₹ 3645 from a shop. How much money he has to pay for both the articles?

Solution: Cost of the washing machine = ₹ 5468
 Cost of the geyser = ₹ 3645
 Total cost of both the articles = ₹ (5468 + 3645)
 = ₹ 9113

Hence, Mr Khanna has to pay ₹ 9113 in all.

1	1	1	← Carry
5	4	6	8
3	6	4	5
9	1	1	3

Example 2: There are 4573 men, 3968 women and 1436 children in a town. What is the population of that town?

Solution: Number of men in the town = 4573
 Number of women in the town = 3968
 Number of children in the town = 1436
 Total population of the town = 4573 + 3968 + 1436
 = 9977

Hence, the total population of the town is 9977.

1	1	1	← Carry
4	5	7	3
3	9	6	8
1	4	3	6
9	9	7	7

Example 3: In a gram panchayat election, there were three candidates. They got 3587 votes, 2874 votes and 708 votes respectively. If 59 votes were found invalid, how many votes were polled in all?

Solution: Number of votes polled to first candidate = 3587
 Number of votes polled to second candidate = 2874
 Number of votes polled to third candidate = 708
 Number of invalid votes = 59
 Total number of votes polled = 3587 + 2874 + 708 + 59
 = 7228

Hence, the number of votes polled is 7228.

Carry →	2	2	2	
	3	5	8	7
	2	8	7	4
		7	0	8
+			5	9
	7	2	2	8



Exercise 9

1. There are 2576 boys and 1894 girls studying in a school.
What is the total number of pupils in the school?
2. Total monthly expenditure in Mr Jain's family is ₹ 7385.
If they save ₹ 1895 per month, what is their monthly income?
3. On Kunal's admission in a school, his father paid ₹ 2758 as fees,
spent ₹ 1367 on his books and ₹ 975 on his school uniform.
How much money did he spend in all?
4. If 4508 men, 2987 women and 1395 children live in a village, what is the total population of the village?
5. In an orchard, there are 1863 coconut trees, 1077 mango trees, 978 tamarind trees and 169 orange trees. How many trees are there in all in the orchard?
6. The number of visitors to a museum on four consecutive days of a week were 2384, 3538, 1835 and 679 respectively. It remained closed for the next three days. What is the total number of visitors during the week?
7. In a school library, there are 1968 books on English, 2056 books on Tamil, 1735 books on Mathematics, 876 books on Science and 1608 books on other subjects. How many books are there in all in the library?
8. A number exceeds 6897 by 2478. What is that number?
9. The difference between two numbers is 1876. If the smaller number is 7948, find the larger number.
10. The cost of a music system is ₹ 6356 more than that of a bicycle. If the cost of the bicycle is ₹ 1765, what is the cost of the music system?
11. The total number of students in primary, middle and senior classes of a school are 5028, 2397 and 1845 respectively. What is the total strength of the school?
12. A poultry farm produces 1639, 2578, 1456 and 995 eggs in four consecutive days. How many eggs altogether are produced in these days?
13. Mr Gupta bought a mixer grinder set for ₹ 3485, a quilt for ₹ 2738 and a bag for ₹ 847. How much had he to pay in all?



**Mental Addition**

You have now learnt to add two or more numbers by writing them columnwise. But, sometimes, in our everyday life, it is not possible to write and add. At such times, one needs to add mentally.

Let us take an example.

A boy is standing at a shop. He buys a pencil box for ₹ 56 and a colour set for ₹ 23. How much money does the boy have to pay to the shopkeeper? Clearly, we need to add 56 and 23.

To add mentally, we think of

56 as 50 and 6 and 23 as 20 and 3

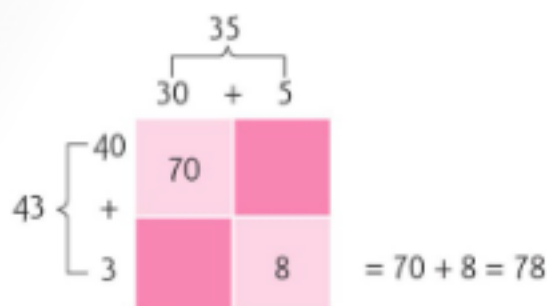
Now, 50 and 20 is 70; 6 and 3 is 9.

70 and 9 is 79.

So, the boy has to pay ₹ 79 to the shopkeeper.

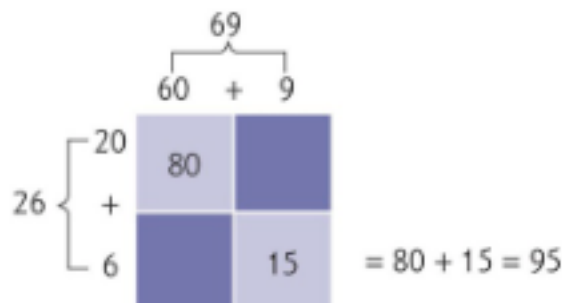
We may, thus, add any two numbers mentally as shown below:

1. $35 + 43$



So, $35 + 43 = 78$.

2. $69 + 26$



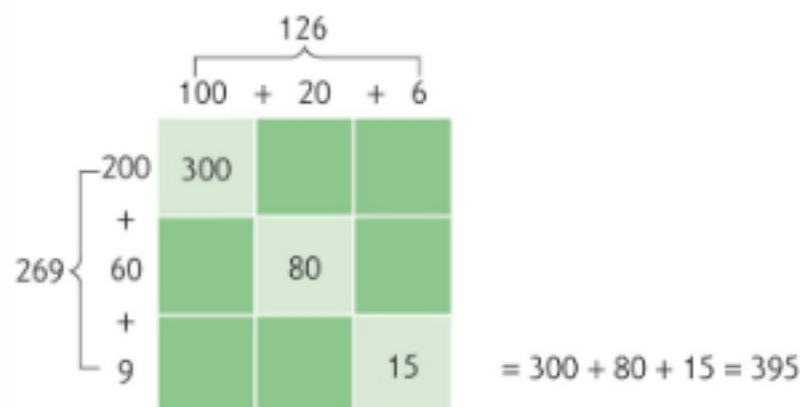
So, $69 + 26 = 95$.

Now, try adding 3-digit numbers in the same manner. Here, for 3-digit numbers, we have to sort out the hundreds, tens and ones and then add them separately.



Consider the following example.

3. $126 + 269$



So, $126 + 269 = 395$.

Add mentally.

- | | | |
|---------------|----------------|----------------|
| 1. $41 + 24$ | 2. $23 + 57$ | 3. $52 + 46$ |
| 4. $37 + 38$ | 5. $69 + 18$ | 6. $42 + 29$ |
| 7. $138 + 44$ | 8. $643 + 345$ | 9. $248 + 193$ |
10. Kirti has 27 stamps with her. Rohit gave her 36 stamps more. How many stamps does Kirti have now?
11. After distributing 58 sweets among the children, Deepak is left with 28 sweets. How many sweets did Deepak have with him at the beginning?
12. A train compartment is carrying 158 people. Another compartment is carrying 229 people. In all, how many people are there in both the compartments?



Things to Remember

1. When some numbers are added, each number is called an **addend**. The result obtained after addition is called their **sum**.
2. The sum of two numbers does not change when the order of addends is changed. This is called the **order property** for addition of numbers.
3. The sum of three or more numbers does not change when their grouping is changed. This is called **grouping property** for addition of numbers.
4. The sum of a number and 0 is equal to the number itself. This is called **additive property** for zero.

**QUESTION BAG 1****(Objective Type Questions)****Tick (✓) the correct answer.**

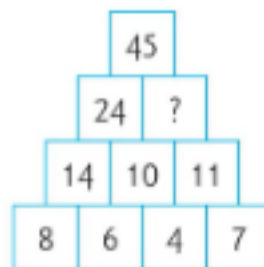
1. $14 + 16 + 12 + 18 = ?$
(a) 40 ☐ (b) 50 ☐ (c) 60 ☐ (d) 70 ☐
2. $25 + 35 + 45 + 55 = ?$
(a) 130 ☐ (b) 140 ☐ (c) 150 ☐ (d) 160 ☐
3. The ones digit in the sum of 3656 and 1948 is
(a) 4 ☐ (b) 6 ☐ (c) 7 ☐ (d) 8 ☐
4. If we add 9 tens, 5 hundreds and 3 ones, then the result is
(a) 17 ☐ (b) 395 ☐ (c) 593 ☐ (d) 953 ☐
5. A school library had 2698 books. The school bought 578 more books. How many books does the library have now?
(a) 3196 ☐ (b) 3256 ☐ (c) 3266 ☐ (d) 3276 ☐
6. Tanya sold 27 chocolates and 47 pastries. Vanya sold 35 chocolates and 39 pastries. How many pastries were sold in all?
(a) 62 ☐ (b) 66 ☐ (c) 82 ☐ (d) 86 ☐
7. There are three primary schools in a city. The total strength of these schools is 1200, 1300 and 1500 respectively. How many pupils are there in all in the three schools?
(a) 3000 ☐ (b) 3500 ☐ (c) 3800 ☐ (d) 4000 ☐
8. Joe has ₹ 38, Anna has ₹ 25 more than Joe. How much money do they have together?
(a) ₹ 63 ☐ (b) ₹ 76 ☐ (c) ₹ 91 ☐ (d) ₹ 101 ☐
9. $40 + 30 + 20 = \boxed{} + 10$
The numeral that should fill the empty box above is
(a) 60 ☐ (b) 70 ☐ (c) 80 ☐ (d) 90 ☐
10. $1345 + 55 + 100 = \boxed{} + 500$
What should come in the empty box above?
(a) 900 ☐ (b) 1000 ☐ (c) 1050 ☐ (d) 1100 ☐

11. Every month Mr Paul spends ₹ 3650 and saves ₹ 5550. What is his monthly income?
 (a) ₹ 8200 ☐ (b) ₹ 9000 ☐ (c) ₹ 9200 ☐ (d) ₹ 9400 ☐
12. In the sum of 1984, 2417 and 1689, how many thousands are there?
 (a) 5 ☐ (b) 6 ☐ (c) 7 ☐ (d) 8 ☐
13. Find the missing number in the following addition:

$$\begin{array}{r}
 \begin{array}{|c|c|c|}
 \hline
 4 & 8 & 9 \\
 \hline
 2 & \square & 7 \\
 \hline
 \end{array} \\
 + \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 7 & 5 & 6 \\
 \hline
 \end{array}
 \end{array}$$



- (a) 4 ☐ (b) 5 ☐ (c) 6 ☐ (d) 7 ☐
14. A town has 4456 adults and 2887 children. The population of the town is
 (a) 6233 ☐ (b) 7233 ☐ (c) 7343 ☐ (d) 7443 ☐
15. Study the pattern and find the missing term.



QUESTION BAG 2

1. Fill in the empty boxes.

- (a) 100 = tens + 10 ones
- (b) 1000 = hundreds + 10 tens
- (c) 200 = hundreds + 10 tens
- (d) 400 = hundreds + 9 tens + ones
- (e) 3000 = hundreds



2. Fill in the empty boxes.

(a) $3256 + 100 =$

(b) $4900 + 1000 =$

(c) $2990 + 10 =$

(d) $9900 + 100 =$

(e) $3060 + 10 =$

(f) $4900 + 100 =$

(g) $6109 +$ $= 6119$

(h) $9929 +$ $= 9930$

(i) $2232 +$ $= 2332$

(j) $6010 +$ $= 6110$

3. Fill in the blanks.

(a) Odd number + Odd number =

(b) Odd number + Even number =

(c) Even number + Even number =

(d) Even number + = Odd number



4. Without actual addition, write if the sum is even or odd.

(a) $6396 + 1048$

(b) $1835 + 2096$

(c) $4464 + 2089$

(d) $6361 + 3187$

5. State whether each of the following statements is true or false.

(a) The sum is always greater than any of the addends.

(b) The digit in the ones place of the sum $3265 + 2857$ is 2.

(c) The sum of any three odd numbers is an even number.

6. Aman's landlord spent ₹ 2868 on repairing the pipelines and ₹ 3595 on electrical fittings in the house. What is the total expenditure of his landlord?

7. A company manufactured 3789 cars in the first year and 4369 cars in the second year. How many cars did the company manufacture in the two years taken together?



8. In a school, there are 1086 students in the primary wing, 878 students in the middle wing and 397 students in the senior wing. How many students study in the school?